MOBILE PENETRATION TEST REPORT GENERATION WITH CVSS 3.1 CALCULATOR

Mobile penetration Testing & Ethical Hacking

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Abstract

Penetration Testing Report is a lightweight web-based application developed to simplify the process of documenting penetration-testing findings and preparing security reports. During a security assessment, testers need to record vulnerabilities, affected components, proof-of-concept steps, remediation measures, evidence, and severity levels in a clear and consistent format. Preparing these details manually can be time-consuming and may lead to inconsistencies in vulnerability scoring and report formatting.

The proposed system provides a single platform where a tester can create a project, add security findings, attach evidence screenshots, and calculate CVSS 3.1 severity scores. The application provides live CVSS score feedback and stores the final calculation along with the finding details. It then uses the stored information to generate a structured PDF penetration-testing report.

The system is implemented using Python, Fast API, SQLite, SQL Alchemy, Jinja2, WeasyPrint, HTML, CSS, Bootstrap, and JavaScript. The developed prototype was tested for project creation, vulnerability entry, CVSS calculation, evidence upload, finding management, and PDF report generation. The project demonstrates how a simple reporting tool can reduce repetitive documentation work and provide a consistent format for penetration-testing reports.

1. Problem Statement

Penetration testing involves identifying security vulnerabilities and documenting important details such as the vulnerability description, affected component, proof-of-concept steps, severity, remediation, and supporting evidence. When these details are recorded manually, the process can become time-consuming and may result in inconsistent report formats or errors in vulnerability severity calculation.

There is a need for a simple and organized system that can bring these activities together in one place. PenReport addresses this problem by providing a web-based platform where testers can create projects, record security findings, calculate CVSS 3.1 scores, attach evidence screenshots, and generate a structured PDF penetration-testing report. This helps make the reporting process more consistent, organized, and efficient.

- Manual CVSS calculation can lead to calculation or rounding mistakes.
- Security reports prepared manually may not follow the same structure every time.
- Evidence screenshots must be collected and inserted into the report separately.
- Students and individual testers may not need a large enterprise reporting platform for a small assessment.

2. Objectives

1. Build a simple browser-based interface for creating penetration-testing projects and findings.
2. Implement CVSS 3.1 Base, Temporal and Environmental metric calculation.
3. Display the CVSS score, severity and vector while the tester is entering a finding.
4. Allow evidence screenshots to be attached to findings.
5. Generate a structured PDF report containing the project summary and technical findings.
6. Provide sample vulnerability data so that the working prototype can be demonstrated quickly.

3. Requirements Analysis

3.1 Functional Requirements

1. Create a new project with project name, client name, tester name, date and scope.
2. Open a project and view its findings.
3. Add, edit and delete security findings.
4. Enter finding title, affected component, description, proof-of-concept steps and remediation.
5. Select CVSS 3.1 metrics and calculate the score.

6. Upload one or more evidence screenshots.
7. Use sample vulnerability autofill for demonstration.
8. Generate the final PDF report from stored project data.

3.2 Non-Functional Requirements

1. The interface should be simple enough for a student tester to use without training.
2. The score calculation should be consistent between the browser and the server.
3. The application should work locally without depending on cloud service.
4. The report generated should be readable and suitable for academic demonstration.
5. The application should give clear visual feedback for vulnerability severity.

3.3 Hardware and Software Requirements

Category	**Requirement**
Processor	Dual-core CPU or better
RAM	Minimum 2 GB
Storage	About 100 MB plus evidence files
Operating System	Windows / Linux / macOS
Language	Python 3.10+
Backend	FastAPI + Uvicorn
Database	SQLite with SQLAlchemy
PDF	WeasyPrint
Frontend	HTML5, Bootstrap 5, JavaScript
Browser	Chrome, Firefox or Edge

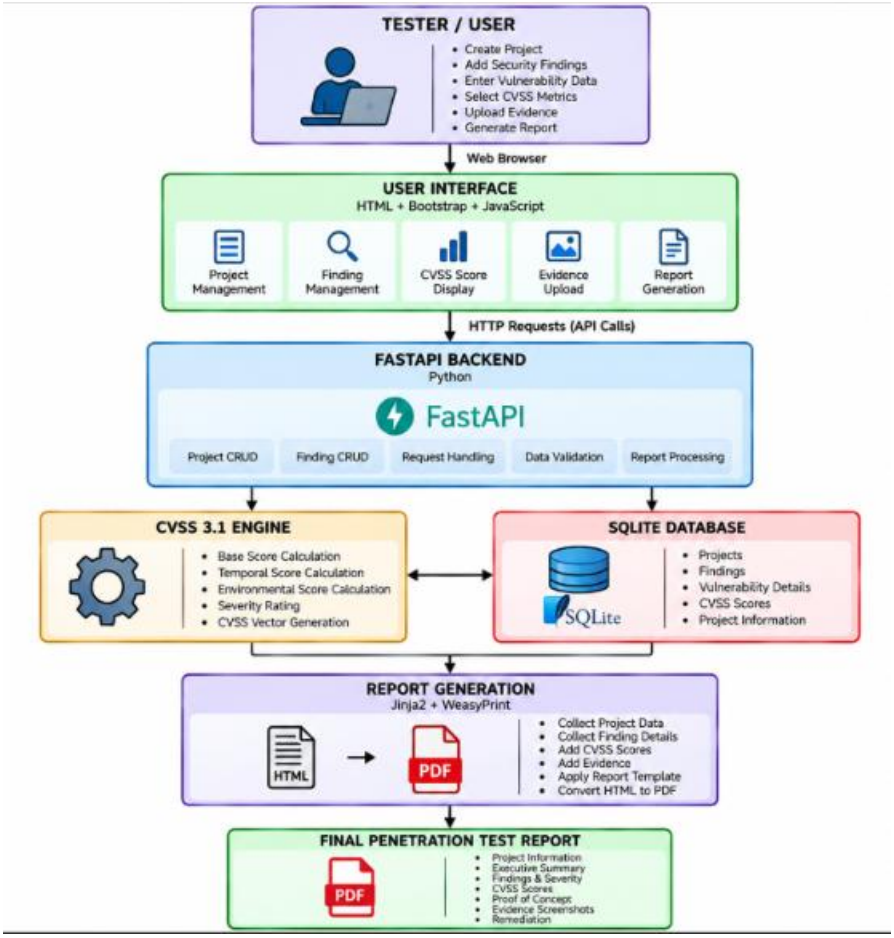
4. Proposed Solution

The proposed solution is **PenReport**, a web-based penetration-testing report management system designed to simplify the process of recording and reporting security vulnerabilities. Instead of maintaining vulnerability details, severity calculations, evidence, and report formatting separately, the system brings these activities together in a single platform.

The tester can first create a project by entering basic details such as the project name, client name, tester name, testing date, and scope. After creating the project, individual security findings can be added with details such as the vulnerability title, affected component, description, proof-of-concept steps, and remediation.

The system includes a **CVSS 3.1 scoring module** that allows the tester to select the required metrics and obtain the vulnerability score and severity. Evidence screenshots can also be attached to individual findings. Once all findings have been entered, PenReport uses the stored project information and findings to generate a structured **PDF penetration-testing report**.

4.1 System Architecture



4.2 Working Flow

- The tester creates a project by entering basic project details.
- Security findings are added with details such as vulnerability, affected component, description, PoC, and remediation.
- The tester selects the required CVSS 3.1 metrics, and the system calculates the score and severity.
- Evidence such as screenshots can be added to the finding.
- The finding details and CVSS score are saved in the database.
- After all findings are completed, the system generates a structured PDF penetration-testing report.

- The final report contains the project details, findings, CVSS scores, evidence, and remediation details.

5. Technologies and Tools Used

Technology / Tool	Use in the Project
Python	Main programming language and application logic
FastAPI	Web backend and request handling
Uvicorn	Local application server
SQLite	Local storage for projects and findings
SQLAlchemy	Database ORM
Jinja2	HTML report templating
WeasyPrint	HTML/CSS to PDF conversion
Bootstrap 5	Responsive user-interface components
JavaScript	Live CVSS calculation and sample autofill
HTML/CSS	Application pages and report layout
CVSS 3.1	Vulnerability severity scoring

6. Implementation

The PenReport describes how the proposed system is developed to manage penetration-testing findings and generate a structured security report. The system follows a simple workflow in which the tester creates a project, enters vulnerability details, calculates the CVSS 3.1 score, adds supporting evidence, and stores the information in the database. Finally, the stored data is used to generate a PDF penetration-testing report.

The system is implemented as a web-based application using Python, FastAPI, SQLite, SQLAlchemy, HTML, Bootstrap, JavaScript, Jinja2, and WeasyPrint. The methodology focuses on keeping the reporting process organized, reducing manual work, and maintaining consistency in vulnerability scoring and report generation.

6.1. CVSS 3.1 Implementation

- **Project Creation** – The tester creates a project and enters basic details such as project name, client name, tester name, date, and scope.

- **Finding Management** – The tester adds individual security findings with details such as vulnerability name, description, affected component, proof of concept, remediation, and evidence.

- **CVSS 3.1 Calculation** – The system provides a CVSS calculator where the tester selects the required metrics. The application calculates the Base, Temporal, and Environmental scores and determines the severity.

- **Database Storage** – Project and vulnerability information is stored in the SQLite database. The system provides functions to create, update, view, and delete projects and findings.

- **Evidence Management** – Screenshots and supporting evidence can be added to individual findings and included in the final report.

- **PDF Report Generation** – The stored project and finding information is passed to the report template. Jinja2 is used for creating the report structure and WeasyPrint converts it into a PDF document.

- **Testing and Validation** – The system is tested by creating projects, adding findings, calculating CVSS scores, managing evidence, and generating PDF reports. Sample vulnerabilities such as SQL Injection, IDOR, XSS, Weak Password Policy, Missing Security Headers, and Sensitive Information Disclosure are also provided for testing.

6.2 Team Contribution

Team Member	Suggested Contribution to Mention
Member 1	Worked on the **implementation of the PenReport project**, including the web application, CVSS 3.1 calculator, database, vulnerability management, and PDF report generation.
Member 2	Prepared the **PowerPoint presentation**, including the project overview, objectives, methodology, system architecture, implementation, results, and future scope.
Member 3	Prepared the **project report**, including the introduction, problem statement, objectives, methodology, system design, implementation details, testing, results, and conclusion.

7. Results

The PenReport project was successfully implemented and tested to verify its main functionalities. The system allows users to create projects, record security findings, calculate CVSS 3.1 scores, add evidence, and generate a structured PDF penetration-testing report. The testing showed that the major features of the application work as expected.

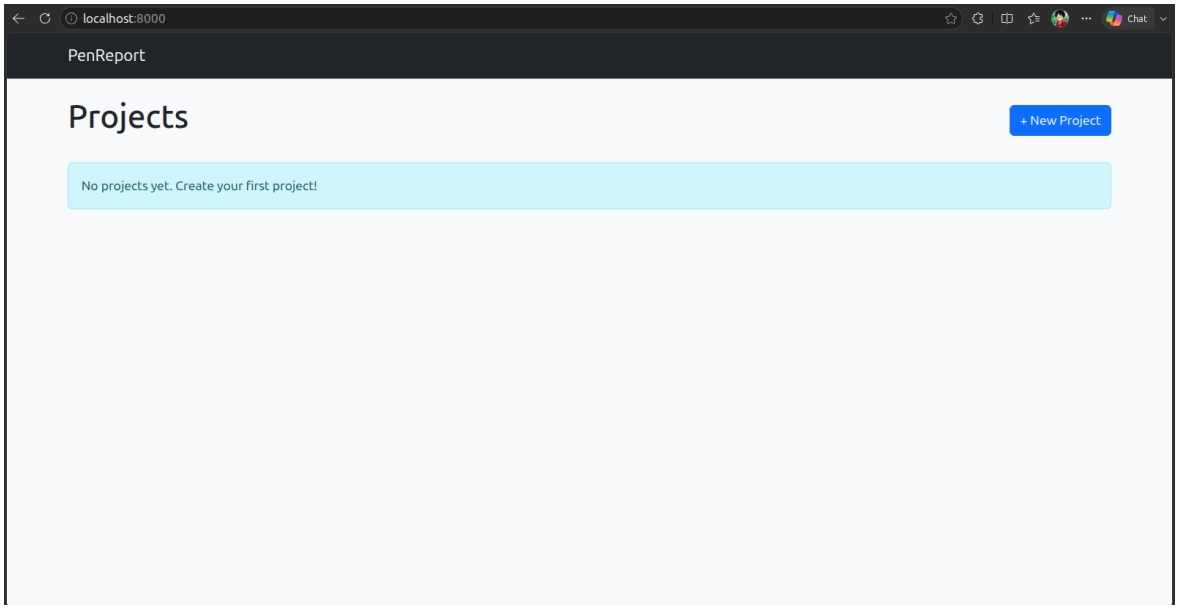


Figure 1. PenReport dashboard before a project is created.

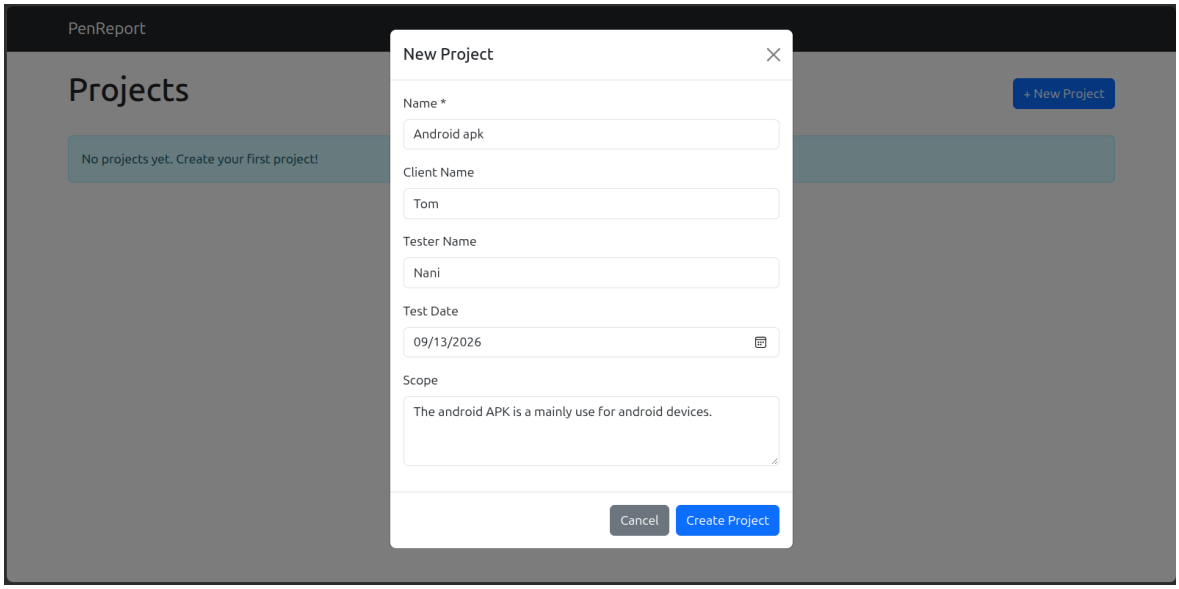


Figure 2. New Project form.

After the project is created, the project page displays the project information and provides buttons for adding findings and generating the PDF. This makes the report-building process more direct than preparing the same information manually.

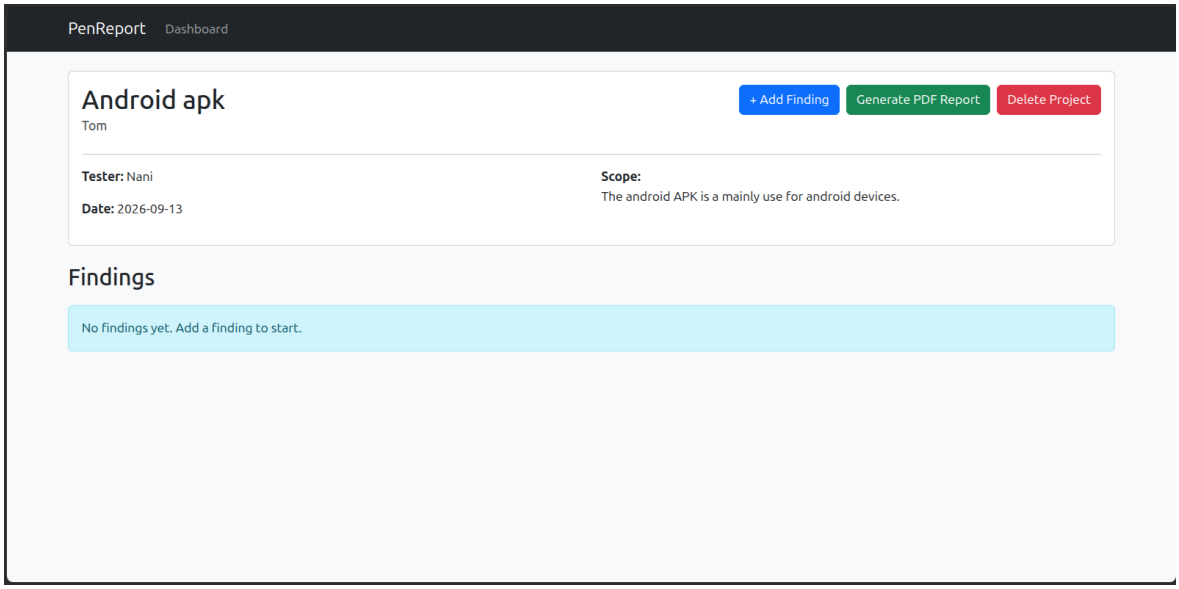


Figure 3. Project details page with the finding area.

The finding screen contains the vulnerability details on the left and the CVSS panel on the right. The supplied demonstration shows Stored Cross-Site Scripting (XSS) as a sample finding, with the affected component, description, proof-of-concept steps and remediation fields.

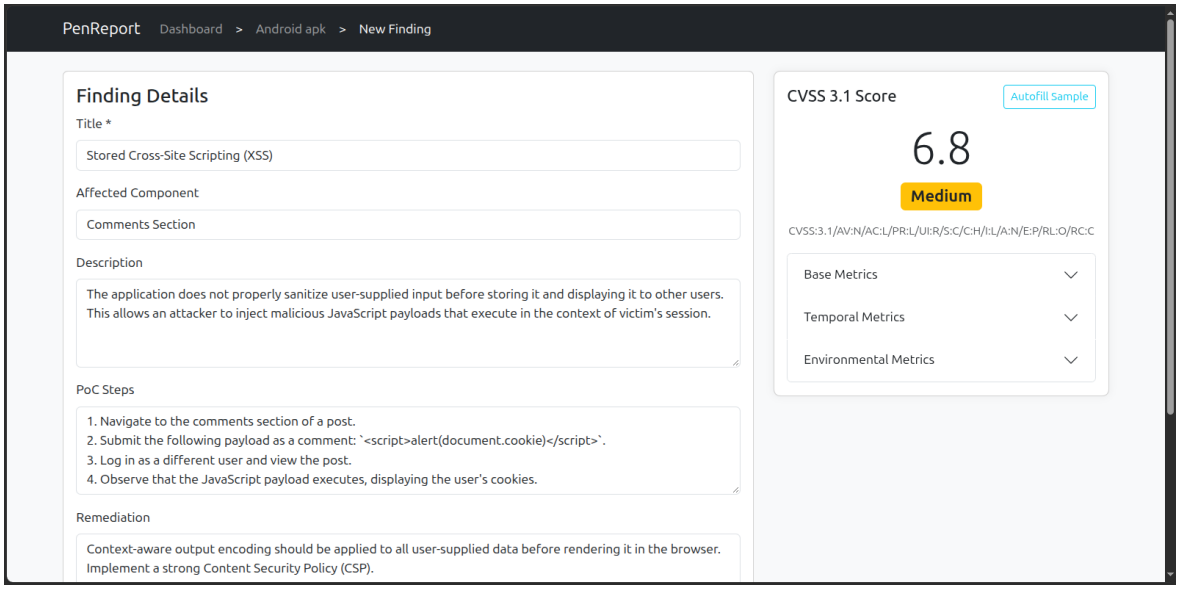


Figure 4. Finding entry screen with CVSS 3.1 panel.

The application also supports evidence screenshots. The supplied evidence preview demonstrates that an image can be attached to a finding and reviewed before the finding is saved.

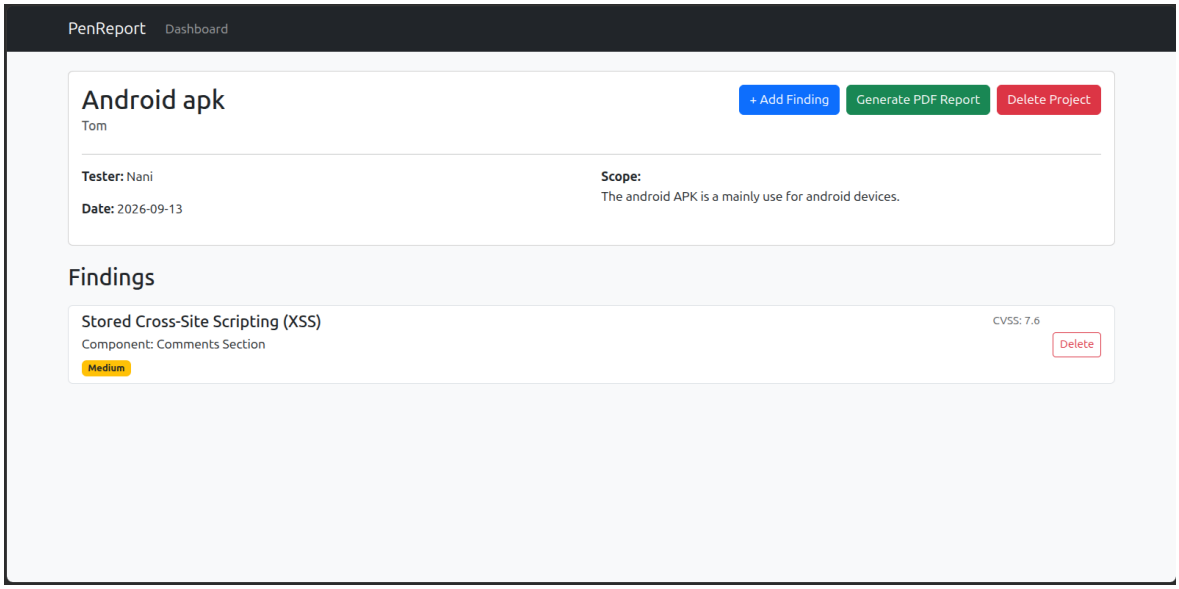


Figure 5. Evidence screenshot preview attached to a finding.

The generated sample penetration test PDF supplied with the project is five pages long. Its executive summary reports two findings, SQL Injection with a Critical severity and CVSS 9.8, and IDOR with High severity and CVSS 8.1. The technical pages contain the affected component, PoC steps and remediation for each finding.

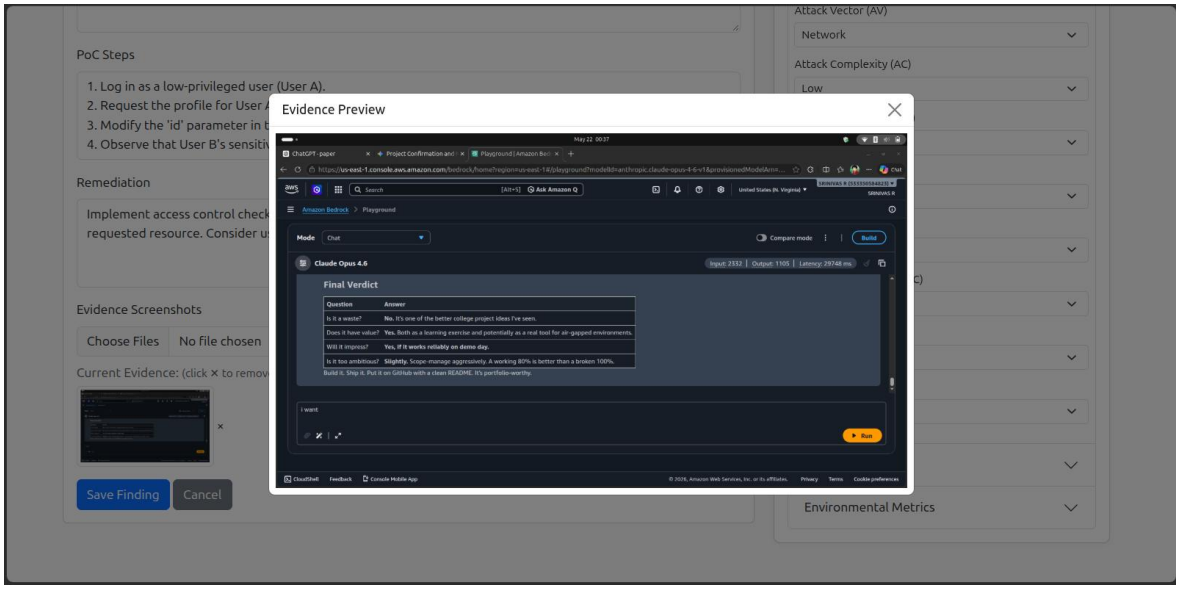


Figure 6. Evidence/report workflow demonstrated in the project.

7.1 Functional Test Results

Test	Expected Result	Observed / Recorded Result
Create project	Project should appear in dashboard	Passed
Open project	Project details should load	Passed
Add finding	Finding should be stored	Passed
Live CVSS scoring	Score and severity should update	Passed
Edit finding	Changes should remain after saving	Passed
Delete finding	Finding should be removed	Passed
Evidence upload	Evidence should be stored and displayed	Passed
Generate PDF	PDF should contain projects and findings	Passed

These functional results are consistent with the testing table in the supplied project documentation, which records successful tests for project creation, finding management, live CVSS scoring, evidence upload and PDF generation.

8. Applications

PenReport can be applied in different areas of cybersecurity and penetration testing. Some of its main applications are:

1. **Penetration Testing Reports** – Helps security testers document vulnerabilities and prepare professional reports.
2. **Mobile Application Security Testing** – Can be used to record and report security findings identified during mobile application testing.
3. **Vulnerability Assessment** – Helps organize vulnerabilities, affected components, severity levels, and remediation details.
4. **CVSS-Based Risk Assessment** – Provides CVSS 3.1 scores to help classify the severity of identified vulnerabilities.
5. **Security Audit Documentation** – Can be used to maintain structured records of security assessment findings and evidence.
6. **Academic and Training Projects** – Useful for students and beginners to practice vulnerability documentation and penetration-testing report preparation.
7. **Security Consultancy** – Can support security professionals in preparing consistent client-facing penetration-testing reports.

9. Challenges and Limitations

Challenges

- Implementing the CVSS 3.1 calculation correctly.
- Managing different vulnerability details and evidence.
- Generating a properly formatted PDF report.

- Keeping the project data organized and consistent.
- Testing all features to ensure they work correctly.

Limitations

- No user authentication is included.
- The system is mainly designed for single-user use.
- Reports can currently be generated only in PDF format.
- No advanced dashboard or analytics are available.
- Data is stored locally, so it is not suitable for multi-user or cloud-based use.

10. Future Scope

- Add user login and authentication.
- Support JSON and CSV report formats**.**
- Add a report preview before generating the PDF.
- Add charts and dashboards for vulnerability analysis.
- Support CVSS 4.0 in the future.
- Add dark mode and improved UI.
- Support cloud-based and multi-user access**.**
- Provide Docker deployment for easier installation.

11. Conclusion

PenReport was successfully developed as a simple web-based tool for penetration-testing report preparation. The system allows testers to record vulnerabilities, calculate CVSS 3.1 scores, add evidence, and generate structured PDF reports.

The project helps reduce manual work and keeps vulnerability information and report formatting organized in one place. It also provides a simple and accessible solution for individual testers and academic projects.

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